



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 311: Computer Architecture

Batch: 22nd

Date: 30/06/2024

Mid-Term Examination

Semester: Spring 2024

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions of the following.

5 × 6 = 30

1. (a) Explain the differences between the Mechanical and Electronic Computer. 3
(b) Define Babbage's Difference Engine. 3
2. (a) Write the limitations of first generation computer. 2
(b) Outline the Structure of Babbage's Analytical Engine. 4
3. (a) Draw the diagram for Memory Hierarchy. 2
(b) Construct single and three level cache organizations with main memory. 4
4. (a) Recall the system management process of Second Generation Computers. 3
(b) Define multichip module, IC density and IC families. 3
5. (a) Draw the flowchart for Cache Read Operation. 3
(b) Explain Cache Addressing and mention the key aspects of Split Caches. 3
6. (a) Construct an IAS program to add two numbers stored in main memory. 3
(b) Analyze the following instruction sets: 3
 - i) $AC := MQ$
 - ii) $AC := AC + M(X)$
 - iii) go to $M(X, 0:19)$
7. Demonstrate the organization of the CPU and main memory of the IAS computer. 6



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 317: Computer Networks

Batch: 22nd

Date: 09/06/2024

Mid-Term Examination

Semester: Spring 2024

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions of the following.

5 × 6 = 30

1. (a) Define Computer Networks. Briefly explain **fault tolerance** and **scalability** of computer networks. 3
(b) Illustrate the function of Data Link Layer with figure. 3
2. (a) Compare TCP/IP and OSI model layers with figure. 3
(b) Examine the differences between hub and switch. 3
3. (a) Interpret the working process of a Modem with diagram. 2
(b) Summarize the functions of OSI Layers with diagram. 4
4. (a) Explain the components of data communications with figure. 3
(b) Define three criteria of a Network. 3
5. (a) Why do we need to do subnetting? What is the default subnet mask for Class B? 3
(b) How can we demonstrate the range of IP address for every classes (**Class A, B and C**), show the steps. 3
6. (a) State three application layer protocols and write their uses. 3
(b) Which layer is called the heart of OSI? Outline the control of this layer over the network. 3
7. Suppose you are an ISP provider. You need to distribute IP among 32 hosts. Now consider this IP (192.168.20.0) for subnetting.
(a) Determine the number of Network ID and Host ID from this scenario 2
(b) Demonstrate the number of valid host with their respective network and broadcast IP in a chart. 4



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 313: Database Management System

Batch: 22nd

Semester: Spring 2024

Date: 11/06/2024

Mid-Term Examination

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions.

Marks: 5 x 6 = 30

1. (a) State the merits and demerits of a flat file model. 2
(b) Explain the ACID property of a DBMS. 4
2. (a) State different types of notation that used in an E-R diagram. 2
(b) Explain the constructing steps of an E-R diagram. 4
3. (a) Define index of a DBMS. 1
(b) Briefly explain the useful and drawbacks of prototyping and draw the flowchart for prototyping during requirement determination of a DBMS. 5
4. (a) Define the degrees of a relationship in an E-R diagram. 2
(b) Briefly explain the Long-held Organizational rules that are being eliminated through Disruptive technologies. 4
5. (a) What is meant by a computerized database? 2
(b) Find the ways that one can change the structure of a table. 4
6. Take a look at the scenario that HUB registrar's office maintains data about the following entities: 6
 - i. Courses, including number, title, credits, syllabus, and prerequisites;
 - ii. Course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom;
 - iii. Students, including std_id, std_name, std_contact, std_email, std_fn, std_mn, std_pr_addr, std_per_addr, std_bg;
 - iv. Instructors, including identification number, name, title, department, rank, contact number, and email address.

Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office. Document all assumptions that you make about the mapping constraints.

7. By using the scenario of Question 6, Implement the SQL command to create all the relations. Write the SQL to create a view name stdvucse containing all the columns from the Students table. 6



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 315 : Microprocessors and Microcontrollers

Batch: 22nd

Semester: Spring 2024

Date: 26/06/2024

Midterm Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

1. (a) What is Microprocessor? Draw the block diagram of a Microprocessor. 2
(b) Write the features of Intel Core i3 Processor. 4
2. (a) Define the following terms: 3×1=3
 - i. Instruction Set
 - ii. Clock Speed
 - iii. Word Length
- (b) Outline the configuration of 8085 Microprocessor. 3
3. (a) Briefly describe the General Purpose Registers in 8085 Microprocessor. 3
(b) Write the comparison between 8085 and 8086 Microprocessor. 3
4. (a) Briefly Explain the functions of SI, SP and IP. 3
(b) What is a Flag Register? Briefly explain the Flag register of 8086 Microprocessor. 3
5. (a) What do you mean by Addressing Modes? Explain the Immediate, Register and Indexed Addressing Mode of 8086 Microprocessor. 3
(b) Suppose, the value of the Code Segment (CS) Register is 10A0H. The values of different offsets are BP: 2025H, IP: 50F0H and SI: 4247H. Now, calculate the effective address of the memory location pointed by the CS register. 3
6. (a) How is the Coprocessor connected to the main processor? 3
(b) Explain the Closely Coupled Multiprocessor configuration with necessary figures. 3
7. Draw and describe the Architecture of an 8086 Microprocessor with necessary indications. 6



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering (CSE)
Department of Computer Science and Engineering (CSE)
Faculty of Science, Engineering and Technology

Course Code: MAT 317

Mid-Term Examination

Date: 6/24/2024

Course Title: Engineering Mathematics

Semester: Spring 2024

Total Time: 1:30 Hours

Total Marks: 30

Answer any Three (03) out of the following questions:

3 × 10 = 30

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

1. (a) What do you mean by Laplace transform? [1]
(b) Find the Laplace Transform of: (i) e^{at} (ii) t (iii) 1 [4]
(c) Solve the differential equation using Laplace transform: [5]
 $\frac{d^2y}{dx^2} - 2\frac{dy}{dt} - 8y = 0$ $y(0) = 3$, $y'(0) = 6$
2. (a) What do you mean by inverse Laplace transform? [1]
(b) Find (i) $L^{-1}\left\{\frac{1}{s^2+6s+13}\right\}$ (ii) $L^{-1}\left\{\frac{6}{s^2-2s-8}\right\}$ [4]
(c) State the convolution theorem. With the help of the convolution theorem find the inverse [5]
Laplace transform of $\frac{1}{s(s^2+1)}$
3. (a) What do you mean by periodic function? Define Fourier series. [2]
(b) Define orthogonal function. Write a set of orthogonal functions. [2]
(c) Find the Fourier series of the function [6]
$$f(x) = \begin{cases} \pi + x, & -\pi < x < 0 \\ 0, & 0 \leq x < \pi, f(x+2\pi) = f(x) \end{cases}$$
4. (a) Find Fourier series expansion of the function $f(x) = x$, $-\pi \leq x \leq \pi$, $f(x+2\pi) = f(x)$ [4]
(b) Find the Fourier sine and cosine series for the function $f(x) = 1$, $0 \leq x \leq 2$ [6]
5. (a) Find the Fourier integral representation of the function $f(x) = \begin{cases} 0, & x < 0 \\ 1, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$ [7]
Also, show that $\int_0^\infty \frac{\sin(x/2)}{x} dx = \frac{\pi}{2}$
- (b) Find the Fourier coefficient for $f(x) = x^2$, $-2 \leq x \leq 2$ [3]



Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 313: Database Management System

Batch: 22nd

Semester: Spring 2024

Date: 28/09/2024

Final Examination

Time: 2 hours

Total Marks: 40

Answer any five questions.

Marks: 5 x 8 = 40

1. (a) Describe the different types of distributed databases with diagram. 4
(b) Determine different types of outer-join for the following relations: 4

People

Name	Age	meal
Bill	22	Hamburger
Sally	28	Pizza
Marry	23	Soft-drinks
Tony	32	Shrimp

Menu

meal	Day
Hamburger	Monday
Pizza	Tuesday
Chicken	Wednesday
Pasta	Thursday
Tacos	Friday

2. Consider the following Employees table:

Emp id	First name	Last name	Email	Job id	Salary	Department id
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- (a) Insert the following data into the above employees table using subquery: 4

Emp id	Last name	Email	Job id	Salary	Department id
99999	Taylor	123@gmail.com	St clerk	50000	50

- (b) Update employee 114's job_id and salary to match that of employee 205 using subquery. 4

3. (a) Using the Employees table of Q2, Show the last name of all employees whose salary is greater than Abel's salary using subquery. 3
(b) Using the Employees table of Q2, Show the last_name, job_id and salary whose job_id containing 141 employee_id not only but also greater than the 143 employee_id's salary. 5

4. (a) Explain the different parts of SQL. 3
(b) Write the SQL command to create the following Student table: 5
Student (Std_id, Std_name, Std_addr, Std_contact, Std_email)

Student

Std id	Std name	Std addr	Std contact	Std e email	Std bg
315181003	Abdul Mannan	Dhaka	01912xxxxxx	abm@gmail.com	AB (+ve)
412163002	Shohagh	Sonargaon	01712xxxxxx	shohagh@gmail.com	A (+ve)
315171013	Tanvir	Daudkandi	01812xxxxxx	Tanvir12@gmail.com	O (+ve)
316123001	Shohagh	Natore	01665xxxxxx	Shohagh1@gmail.com	B (+ve)
412163001	Ramjan	Modonpur	01774xxxxxx	ramjan@gmail.com	O (-ve)

Write the SQL command to insert the data "Std_dob=13-Nov-2018" into the above table.

5. (a) Explain Normalization with an example. 4
(b) State the similarities and dissimilarities between 3NF and BCNF. 4
6. (a) Explain the union and Cartesian product-operations with example. 2



- (b) By using the Student table of Q4, Find the SQL and Relational Algebra for the 3x2=6 followings:

i)	Select only those students who are neither in the CSE department nor in EEE.
ii)	Choosing the id, name, email, and phone number of students whose blood group not only ('O +ve') but also lives in Dhaka.
iii)	Show the id, name email, pb_mrk and sum of marks of the students who are in the CSE department.

7. Consider the following Order table with variable repeating groups. Find 1NF, 2NF and 3NF for 8 the given order table.

Order

Order id	Order date	Customer	Customer email	Grade	Credit points	Item name	Qty reqd	Unit price
1	11/3/98	Joe Smith	joe@iewr.com	A	1	Hammer	5	\$25
1	11/3/98	Joe Smith	joe@iewr.com	A	1	Saw	10	\$30
1	11/3/98	Joe Smith	joe@iewr.com	A	1	Nails	15	\$40
2	12/3/98	Sam	sam@ter.com	B	4	Saw	20	\$30
2	12/3/98	Sam	sam@ter.com	B	4	Nails	25	\$40



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 311: Computer Architecture

Batch: 22nd

Semester: Spring 2024

Date: 30/09/2024

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

1. (a) What is a virtual memory? 2
(b) Explain the working process of a virtual memory with proper diagram. 3
(c) Why it is necessary to use a virtual memory in computer system? 3
2. (a) Explain the function and units of processor level in system design methodology. 4
(b) Mention the components, IC density, and information units of different gates level. 4
3. (a) What are the purposes of addressing mode? 2
(b) Explain 2 types of instruction format with figure and example. 3
(c) Show the step by step working procedure of instruction format. 3
4. Explain the following terms in details from the perspective of computer system design methodology: 8
 - i. System Design
 - ii. HDL
 - iii. System Bus
 - iv. Interrupt
5. (a) What is Bus arbitration? Explain independent Requesting process with figure. 4
(b) A process having 16 address line and it is byte addressable. You need to interface 4096x8 RAM with this processor. Now try to find the following terms: 4
 - i. How many address line are required for the RAM?
 - ii. How many RAM chips are required for this interfacing?
6. (a) Write the differences between RISC and CISC. 2
(b) How interrupt is handled in CISC system? Write the key points only. 2
(c) What is data hazard? Explain the types of data hazard. 4
7. (a) Write the whole process of instruction cycle with necessary diagram. 4
(b) What is Pipelining? Briefly discuss simplified and expanded view of instruction pipeline. 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 315 : Microprocessors and Microcontrollers

Batch: 22st

Semester: Spring 2024

Date: 05/09/2024

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions.

5 × 8 = 40

1. ✓ (a) What is a Microcontroller? Mention the structural components of a Microcontroller. 2
(b) Describe the different types of Microcontrollers with necessary figures. 6
2. ✓ (a) What are the Special Microcontroller features of PIC16F877A? 4
(b) Differentiate between Harvard and Von Neumann Architectures. 4
3. ✓ (a) Briefly describe the CCP module of PIC16F877A. 3
(b) Differentiate between Microcontroller and Microprocessor. 5
4. ✓ (a) State the steps for EEPROM write in PIC16F877A memory interfacing. 4
(b) Define Maskable and Non-Maskable interrupts. Illustrate the Interrupt Cycle with proper indications. 4
5. (a) Explain Interrupt Latency. 3
(b) How does the processor manage simultaneous interrupt requests from multiple devices? Explain. 5
6. (a) What is an RTOS and why is it used? 4
(b) Define Direct Memory Access (DMA). Briefly describe different modes of the 8237 DMA controller. 4
7. ✓ (a) Define Sensor. Explain the role of Transducer with necessary figure. 4
(b) Write the flowchart for a procedure to detect, de-bounce and produce the hex code for a pressed key. 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 317: Computer Networks

Batch: 22nd

Semester: Spring 2024

Date: 25/09/2024

Final Examination

Time: 2 Hour

Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

- ✓ (a) What are the Features of OSPF and how does it work? 4
- (b) Explain the operational states and types of link in OSPF. 4
2. ✓ (a) What is the name of network protocol that operates on MAC layer? 2
- (b) State the algorithms and flowchart of CSMA/CA. 6
3. Consider you are given the address space 172.16.10.0/23 for Sales 1 and Service Department and 192.168.1.0/24 for rest of the Departments. Now you are to subnet the addresses according to the host requirements for different departments that are given in the following table. Then fill up the empty columns with values resulting from your calculation. Use the smallest subnet sizes that will accommodate the relevant number of hosts and do not leave any gap (at the start of given address space or between the subnet).

Department	No. of hosts required	Network address	Broadcast address	Subnet mask
Sales 1	90			
Service	70			
Accounts	20			
Sales 2	12			
HR	10			

4. ✓ (a) Show the ICMP message format with proper diagram. 3
- (b) Suppose you are asked to send a text message using the latest version of OSPF. Your IP is 192.168.10.3. During the process an error is found on area identification. Now show the message format for this scenario. 5
5. ✓ Consider the address 168.11.2.0; we want to create four subnets under this IP. 8
- Now try to find the answer these following:
- Find the CIDR value for 4 subnets.
 - What is the new subnet mask?
 - How many useable host ID for each new subnet?
 - Find out each network ID, usable host ID range and each broadcast ID.

- ~~6.~~ (a) Explain the Client-Server architecture via a simple example. 4
(b) From the idea of socket-based communication how can you establish a path between client and server? 4

7. (a) Define the following terms in short. 4
i. IGRP
ii. ICMP
iii. CSMA/CD
iv. CSMA/CA
(b) State the goals and functions of IGRP for any new modification of network format. 4

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Department of Computer Science and Engineering (CSE)
Faculty of Science, Engineering and Technology

Course Code: MAT 317

Final Examination

Date: 10/2/2024

Course Title: Engineering Mathematics

Semester: Spring 2024

Total Time: 02 Hours

Total Marks: 40

Answer any four (04) of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

1. (a) Write the Legendre differential equation. [1]

(b) Show that $(1 - 2xz + z^2)^{\frac{1}{2}} = \sum_{n=0}^{\infty} z^n P_n(x)$ considering $|x| \leq 1, |z| \leq 1$. Where $P_n(x)$ is the Legendre polynomial of first kind. [6]

(c) Using the generating function of $P_n(x)$, find [3]

$$P_0(x) = 1, P_1(x) = x, P_2(x) = \frac{1}{2}(3x^2 - 1) \text{ and } P_3(x) = \frac{1}{3}(5x^3 - 3x)$$

2. (a) Prove that [10]

i. $\int_{-1}^1 P_m(x) P_n(x) dx = 0$ if $m \neq n$

ii. $\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$ if $m = n$

Where $P_n(x)$ is the Legendre polynomial of first kind.

3. (a) Prove that $e^{\frac{x}{2}\left(\frac{z-1}{z}\right)} = \sum_{n=-\infty}^{\infty} J_n(x) z^n$, Where $J_n(x)$ is the Bessel polynomial. [5]

(b) Find $J_0(x)$ and $J_1(x)$ using the definition of $J_n(x)$. [2]

(c) Prove that $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$. [3]

4. (a) Prove that [10]

i. $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$

ii. $\frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$

iii. $J_n'(x) = J_{n-1}(x) - \frac{x}{n} J_n(x)$

5. (a) If n is any integer then prove $J_{-n}(x) = (-1)^n J_n(x)$ [4]

(b) Prove that $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$ [6]



6. (a) Define Sturm-Liouville problem. [1]

(b) What do you mean by Eigen value and Eigen function? [2]

(c) Find the Eigen value and Eigen function for the boundary value problem $\frac{d^2 y}{dx^2} + \lambda y = 0$; $y(0) = 0$ and $y(\pi) = 0$ [7]